

Lab Assessment Rubrics of Digital System Design (CPE-344)

Lab Number		05								
Lab Title		Introduction to Behavioral Modeling and Implementation of Different Multiplexers in Behavioral Modeling								
Submitted by										
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Rubrics name & number								Marks		
								In-Lab	Post-Lab	
Engineering Knowledge	R2: Use of Engineering Knowledge and follow Experiment Procedures: Ability to follow experimental procedures, control variables, and record procedural steps on lab report.									
	R3: Interpretation of Subject Knowledge: Ability to interpret and explain mathematical and/or visual forms, including equations, diagrams, graphics, figures, and tables.									
Problem Analysis	R6: Experimental Data Analysis: Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.									
Design	R7: Implementing Design Strategy: Ability to execute a solution taking into consideration design requirements and pertinent contextual elements. [Block Diagram/Flow chart/Circuit Diagram]									
	R8: Best Coding Standards: Ability to follow the coding standards and programming practices.									
Modern Tools Usage	R9: Understand Tools: Ability to describe and explain the principles behind and applicability of engineering tools.									
	R11: Tools Evaluation: Ability to simulate the experiment and them using hardware tools to verify The results.									
Individual and Teamwork	R12: Individual Work Contributions: Ability to carry out individual responsibilities.									
	R13: Management of Teamwork: Ability to appreciate, understand and work with multidisciplinary team members.									
Rubrics to follow										
Rubrics	R2	R3	R5	R6	R7	R8	R9	R11	R12	R13
In-Lab										
Post-Lab										

Lab: 05

In Lab Tasks:

Task#01:

1. MUX 4x1

Verilog Code

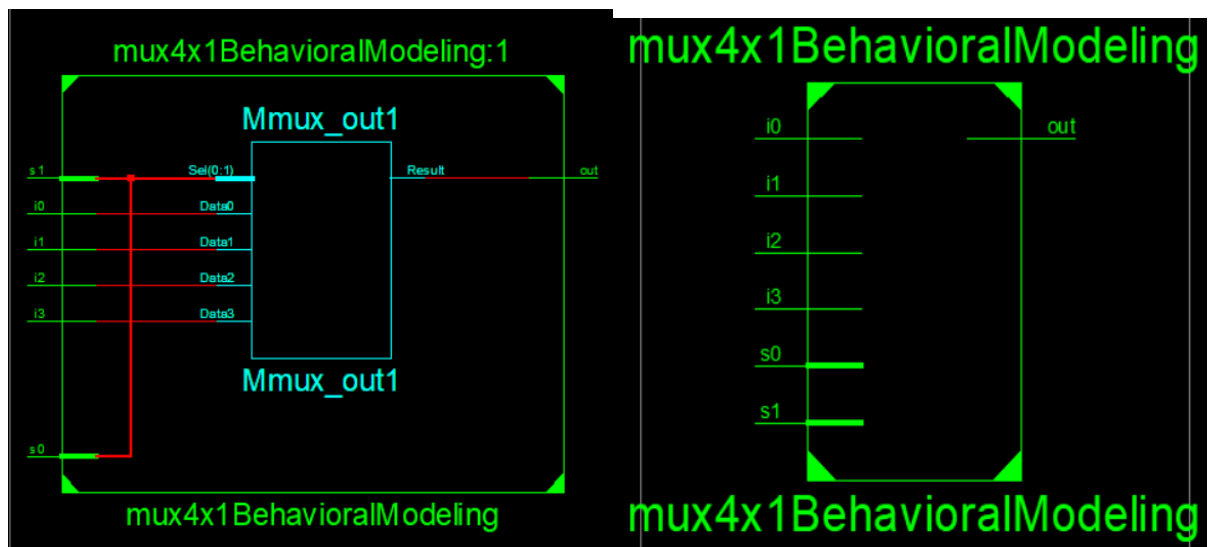
```
module mux4x1BehavioralModeling(out, i3, i2, i1, i0, s1, s0);
input wire i3, i2, i1, i0, s1, s0;
output reg out;

always @(*)
begin
case({s0, s1})
2'b00: out<=i0; 2'b01: out<=i1;
2'b10: out<=i2; 2'b11: out<=i3;
endcase
end
endmodule
```

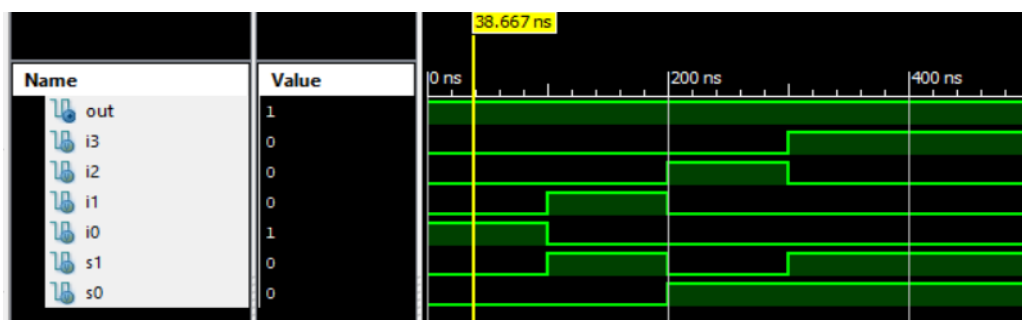
Test Bench

```
module mux4x1BehavioralModelingTestBench;
reg i3, i2, i1, i0, s1, s0;
wire out;
mux4x1BehavioralModeling uut (
.out(out), .i3(i3),
.i2(i2), .i1(i1),
.i0(i0), .s1(s1),
.s0(s0));
initial begin
i3 = 0; i2 = 0;
i1 = 0; i0 = 1;
s1 = 0; s0 = 0;
#100;
i3 = 0; i2 = 0; i1 = 1;
i0 = 0; s1 = 1; s0 = 0;
#100;
i3 = 0; i2 = 1; i1 = 0;
i0 = 0; s1 = 0; s0 = 1;
#100;
i3 = 1; i2 = 0; i1 = 0;
i0 = 0; s1 = 1; s0 = 1;
#100;
end
endmodule
```

RTL Schematic



Waveform



2. MUX 8x1

Verilog Code

```
module mux8x1BehavioralModeling(out,i7, i6, i5, i4, i3, i2, i1, i0, s2, s1, s0);
input wire i7, i6, i5, i4, i3, i2, i1, i0, s2, s1, s0;
output reg out;
always @(*)
begin
case({s0, s1, s2})
3'b000: out<=i0; 3'b001: out<=i1;
3'b010: out<=i2; 3'b011: out<=i3;
3'b100: out<=i4; 3'b101: out<=i5;
3'b110: out<=i6; 3'b111: out<=i7;
endcase
end
endmodule
```

Test Bench

```
module mux8x1BehavioralModelingTestBench;
reg i7, i6, i5, i4, i3;
reg i2, i1, i0, s2, s1, s0;
wire out;
```

```
mux8x1BehavioralModeling uut (
```

```
.out(out),
```

```
.i7(i7),.i6(i6),.i5(i5),
```

```
.i4(i4),.i3(i3),.i2(i2),
```

```
.i1(i1),.i0(i0),.s2(s2),
```

```
.s1(s1),.s0(s0));
```

```
initial begin
```

```
i7 = 0; i6 = 0; i5 = 0;
```

```
i4 = 0; i3 = 0; i2 = 0;
```

```
i1 = 0; i0 = 1; s2 = 0;
```

```
s1 = 0; s0 = 0;
```

```
#100;
```

```
i7 = 0; i6 = 0; i5 = 0;
```

```
i4 = 0; i3 = 0; i2 = 0;
```

```
i1 = 1; i0 = 0; s2 = 0;
```

```
s1 = 0; s0 = 1;
```

```
#100;
```

```
i7 = 0; i6 = 0; i5 = 0;
```

```
i4 = 0; i3 = 0; i2 = 1;
```

```
i1 = 0; i0 = 0; s2 = 0;
```

```
s1 = 1; s0 = 0;
```

```
#100;
```

```
i7 = 0; i6 = 0; i5 = 0;
```

```
i4 = 0; i3 = 1; i2 = 0;
```

```
i1 = 0; i0 = 0; s2 = 0;
```

```
s1 = 1; s0 = 1;
```

```
#100;
```

```
i7 = 0; i6 = 0; i5 = 0;
```

```
i4 = 1; i3 = 0; i2 = 0;
```

```
i1 = 0; i0 = 0; s2 = 1;
```

```
s1 = 0; s0 = 0;
```

```
#100;
```

```
i7 = 0; i6 = 0; i5 = 1;
```

```
i4 = 0; i3 = 0; i2 = 0;
```

```
i1 = 0; i0 = 0; s2 = 1;
```

```
s1 = 0; s0 = 1;
```

```
#100;
```

```
i7 = 0; i6 = 1; i5 = 0;
```

```
i4 = 0; i3 = 0; i2 = 0;
```

```
i1 = 0; i0 = 0; s2 = 1;
```

```
s1 = 1; s0 = 0;
```

```
#100;
```

```
i7 = 1; i6 = 0; i5 = 0;
```

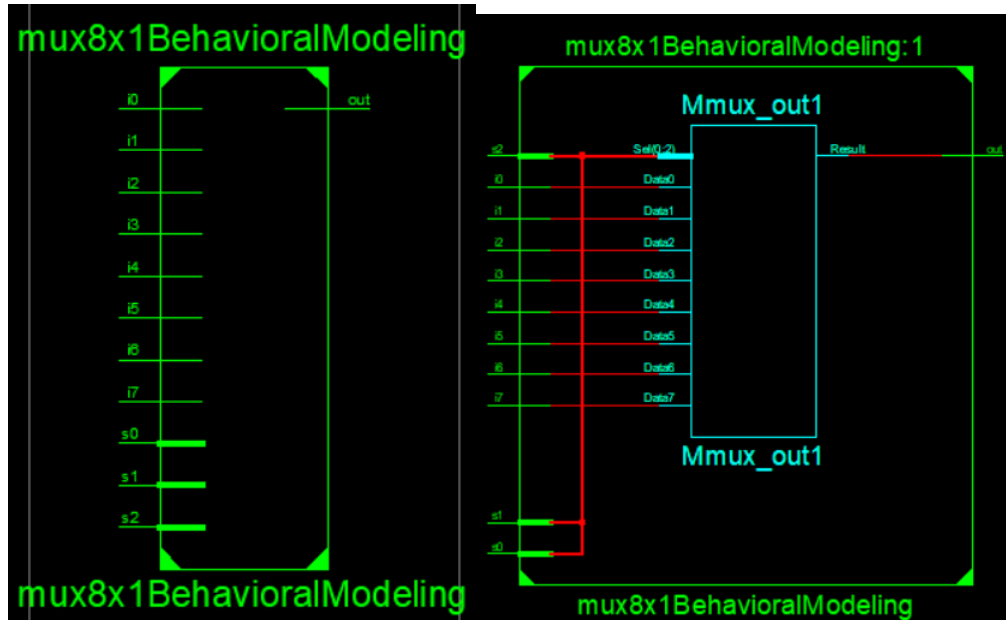
```
i4 = 0; i3 = 0; i2 = 0;
```

```
i1 = 0; i0 = 0; s2 = 1;
```

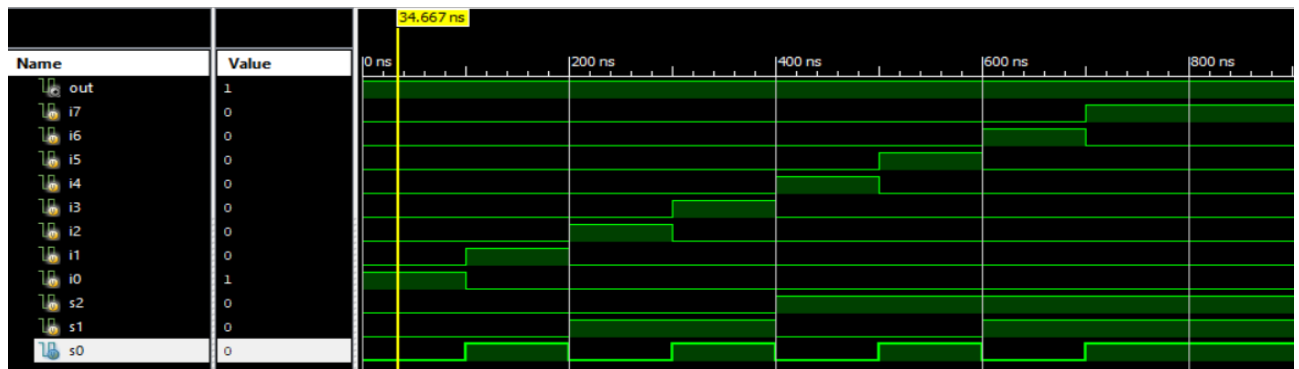
```
s1 = 1; s0 = 1;
```

```
#100;
end
endmodule
```

RTL Schematic



Waveform



Task#02:

Verilog Code

```
module mux4x1UsingifElse(out, i3, i2, i1, i0, s);
input wire i3, i2, i1, i0;
input wire [1:0]s;
output reg out;
always @(*)
begin
if(s==2'b00)
out=i0;
else if(s==2'b01)
out=i1;
else if(s==2'b10)
out=i2;
else if(s==2'b11)
out=i3;
end
endmodule
```

Test Bench

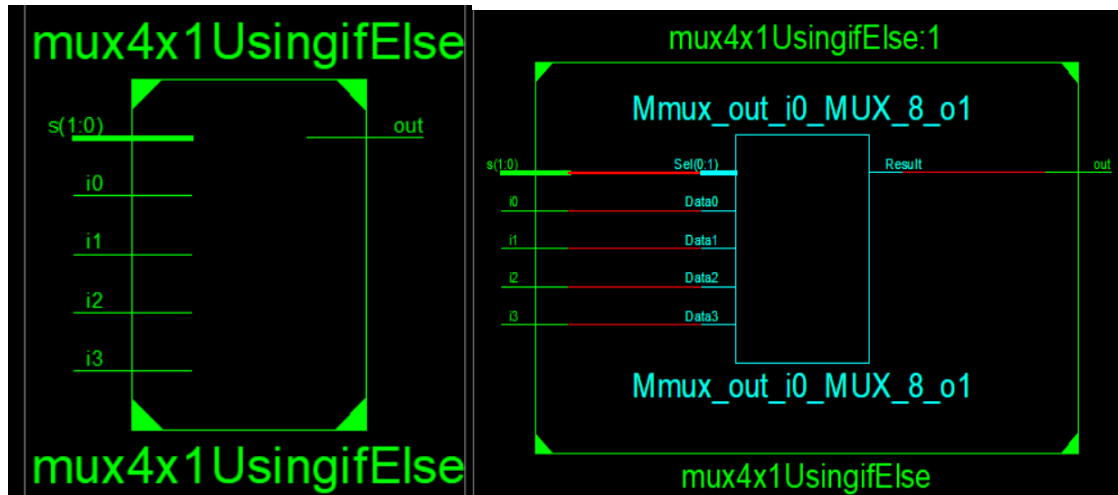
```
module mux4x1ifElseTestBeench;
reg i3, i2, i1, i0;
reg [1:0] s;
wire out;
mux4x1UsingifElse uut (
.out(out),
.i3(i3), .i2(i2),
.i1(i1), .i0(i0),
.s(s));
initial begin
i3 = 0; i2 = 0;
i1 = 0; i0 = 1;
s = 2'b00;
#100;
i3 = 0; i2 = 0;
i1 = 1; i0 = 0;
s = 2'b01;
#100;
i3 = 0; i2 = 1;
i1 = 0; i0 = 0;
s = 2'b10;
#100;
i3 = 1;
i2 = 0;
i1 = 0;
i0 = 0;
```

```

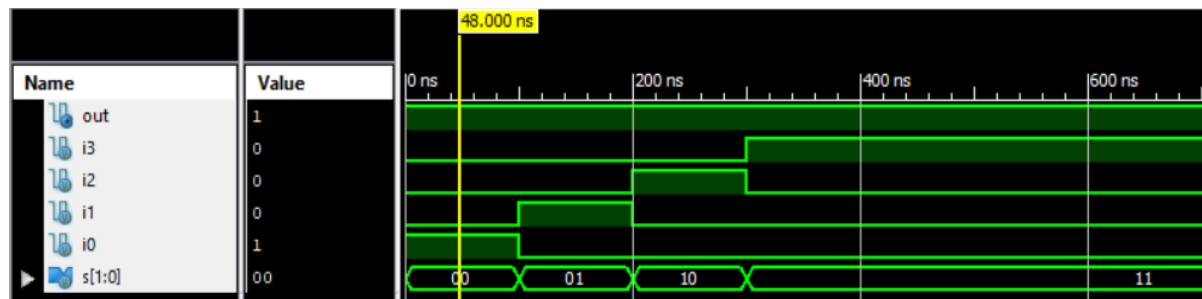
s = 2'b11;
#100;
end
endmodule

```

RTL Schematic



Waveform



MUX 8x1 Using if else Statement

Verilog Code

```
module mux8x1ifElse(out,i7, i6, i5, i4, i3, i2, i1, i0, s);
input wire i7, i6, i5, i4, i3, i2, i1, i0;
input wire [2:0]s;
output reg out;
always@(*)
begin
if(s==3'b000) out = i0;
else if(s==3'b001) out = i1;
else if(s==3'b010) out = i2;
else if(s==3'b011) out = i3;
else if(s==3'b100) out = i4;
else if(s==3'b101) out = i5;
else if(s==3'b110) out = i6;
else if(s==3'b111) out = i7;
end
endmodule
```

Test Bench

```
module mux8x1ifElseTestBench;
reg i7, i6, i5, i4, i3, i2, i1, i0, [2:0]s ;
wire out;
mux8x1ifElse uut (
.out(out), .i7(i7),
.i6(i6), .i5(i5), .i4(i4),
.i3(i3), .i2(i2), .i1(i1),
.i0(i0), .s(s));
initial begin
i7 = 0; i6 = 0; i5 = 0;
i4 = 0; i3 = 0; i2 = 0;
i1 = 0; i0 = 1; s = 3'b000;
#100;
i7 = 0; i6 = 0; i5 = 0;
i4 = 0; i3 = 0; i2 = 0;
i1 = 1; i0 = 0; s = 3'b001;
#100;
i7 = 0; i6 = 0; i5 = 0;
i4 = 0; i3 = 0; i2 = 1;
i1 = 0; i0 = 0; s = 3'b010;
#100;
i7 = 0; i6 = 0; i5 = 0;
i4 = 0; i3 = 1; i2 = 0;
i1 = 0; i0 = 0; s = 3'b011;
#100;
```

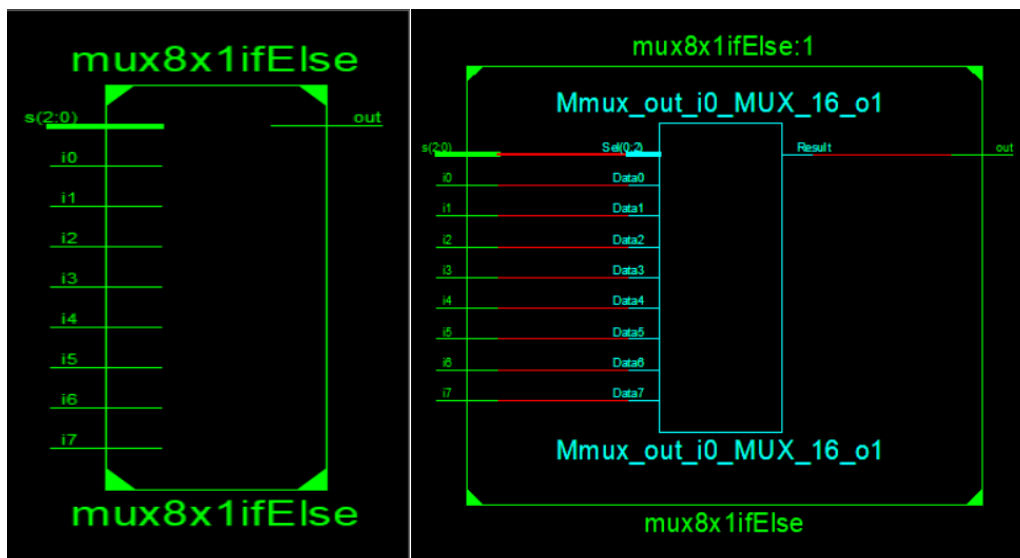


```

i7 = 0; i6 = 0; i5 = 0;
i4 = 1; i3 = 0; i2 = 0;
i1 = 0; i0 = 0; s = 3'b100;
#100;
i7 = 0; i6 = 0; i5 = 1;
i4 = 0; i3 = 0; i2 = 0;
i1 = 0; i0 = 0; s = 3'b101;
#100;
i7 = 0; i6 = 1; i5 = 0;
i4 = 0; i3 = 0; i2 = 0;
i1 = 0; i0 = 0; s = 3'b110;
#100;
i7 = 1; i6 = 0; i5 = 0;
i4 = 0; i3 = 0; i2 = 0;
i1 = 0; i0 = 0; s = 3'b111;
#100;
end
endmodule

```

RTL Schematic



Waveform

